B.Tech Grade Predictor & Goal Setter

Course: Fundamentals of AI and ML (BYOP)

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Project Report

1. Introduction

The current B.Tech curriculum can be confusing for students when it comes to understanding their grades because the internal weightage system is complex. This project proposes a Decision Support Tool that helps students understand their current academic standing. The tool uses a mathematical model to determine how many marks a student needs in the End-term exam to achieve a desired grade.

2. Problem Statement

Many students struggle to understand how Mid-term and Assignment scores influence their final grade. The Mid-term exam is out of 50 marks and Assignments are out of 40 marks, while the final course grade is calculated out of 100. This project performs the required calculations automatically so that students can clearly understand their academic position and plan their goals accordingly.

3. Mathematical Methodology

The tool uses a Weighted Linear Model based on the university's grading rules. Mid-Term: 50 marks with 30% weightage calculated as (Marks / 50) × 30. Assignments: 40 marks with 40% weightage using a direct mapping where marks correspond to percentage contribution. End-Term: 100 marks with 30% weightage. The required marks are calculated using the formula (Target − Current) / 0.3 to determine the score needed in the End-term examination.

4. Implementation Details

The tool is implemented using Python 3.x and runs inside the VS Code terminal. It follows a step-by-step logical structure to ensure accurate calculations. The user inputs Mid-term marks,

Assignment marks, and their target grade. The program then calculates the marks required in the End-term exam and also checks whether the target goal is achievable or impossible.

5. Challenges Faced

One challenge in the project was properly scaling the Mid-term marks to match the 30% weightage while maintaining decimal accuracy. Another challenge was designing the goal-setting logic so that the output provides meaningful and clear feedback to the user based on the calculated results.

6. Logic Enhancements"

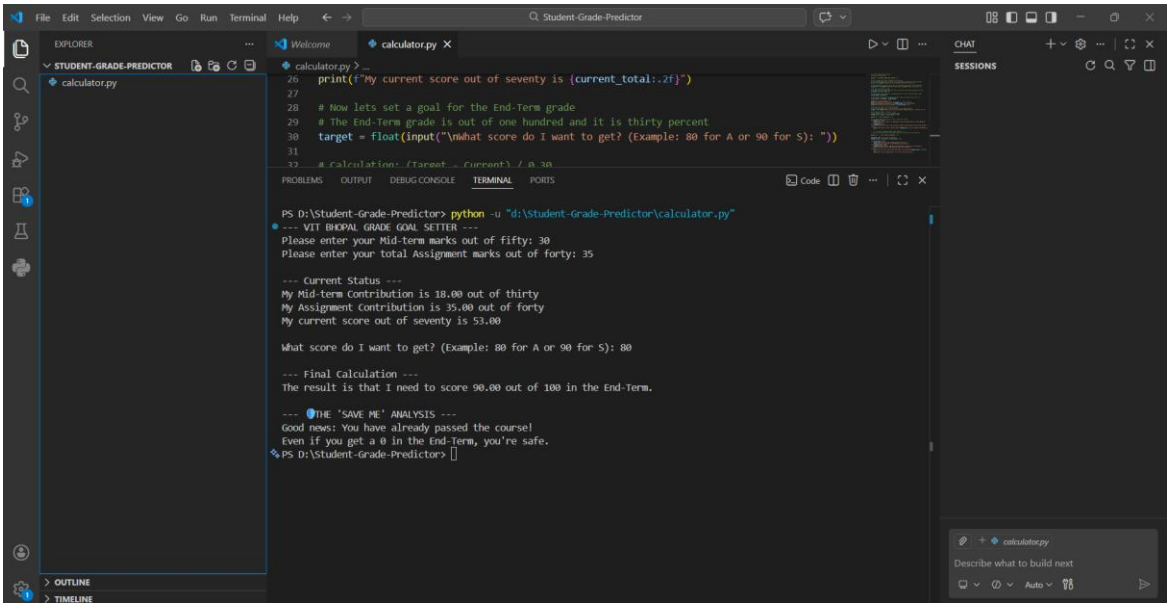
Real-World Utility: "I added a 'Save Me' analysis to help students identify their absolute minimum requirement for passing. This adds a layer of risk management to the tool".

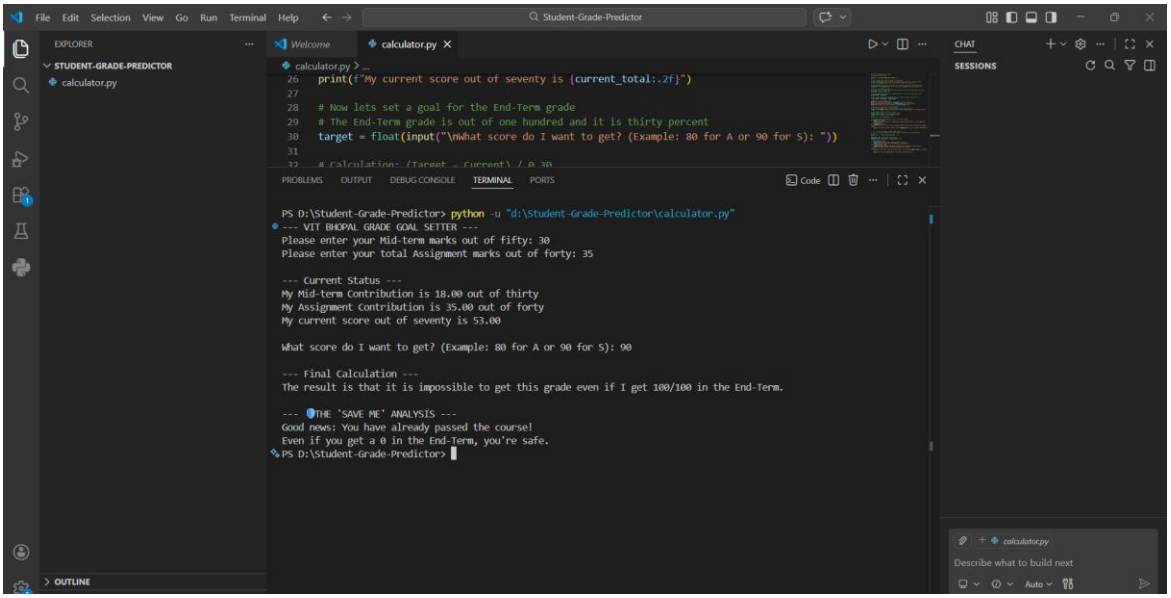
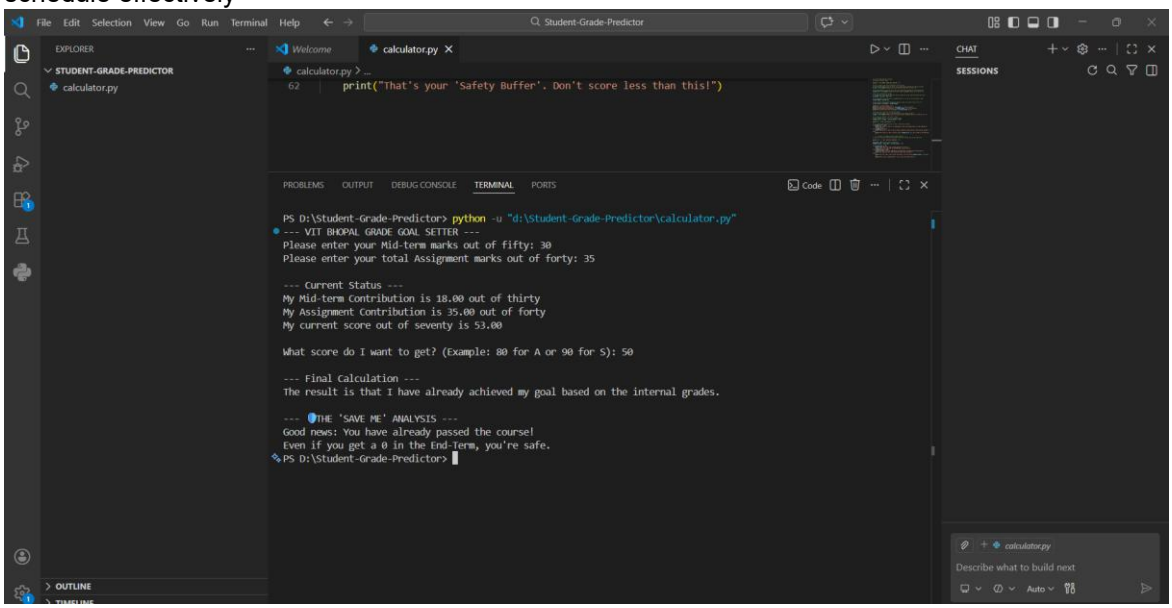
User-Centric Design: "The output uses relatable language to make the academic data easier to process during high-stress exam periods".

Mathematical Edge-Cases: "The logic accounts for students who have already secured enough internal marks (MT + Assignments) to pass the course before the final exam even starts"

Conclusion

This project demonstrates that even simple mathematical models can be used to predict possible outcomes using current academic data. The concept relates to ideas used in supervised learning and predictive analytics. By combining the B.Tech grading structure with a decision support tool, students can better understand their grades and set realistic academic goals. By adding the Safety Buffer feature, the project transitions from a basic calculator to a **Predictive Academic Dashboard**. It doesn't just show a number; it provides a 'Safe Zone' for students to plan their study.


File Edit Selection View Go Run Terminal Help
Student-Grade-Predictor
EXPLORER
STUDENT-GRADE-PREDICTOR
calculator.py
Welcome
calculator.py
calculator.py > ...
print(f"My current score out of seventy is {current_total:.2f}")
# Now lets set a goal for the End-Term grade
# The End-Term grade is out of one hundred and it is thirty percent
target = float(input("\nWhat score do I want to get? (Example: 80 for A or 90 for S): "))
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
Code
PS D:\Student-Grade-Predictor> python -u "d:\Student-Grade-Predictor\calculator.py"
--- VIT BHOPAL GRADE GOAL SETTER ---
Please enter your Mid-term marks out of fifty: 30
Please enter your total Assignment marks out of forty: 35
--- Current Status ---
My Mid-term Contribution is 18.00 out of thirty
My Assignment Contribution is 35.00 out of forty
My current score out of seventy is 53.00
What score do I want to get? (Example: 80 for A or 90 for S): 80
--- Final Calculation ---
The result is that I need to score 90.00 out of 100 in the End-Term.
--- THE 'SAVE ME' ANALYSIS ---
Good news: You have already passed the course!
Even if you get a 0 in the End-Term, you're safe.
PS D:\Student-Grade-Predictor>
OUTLINE
TIMELINE
CHAT
SESSIONS
calculator.py
Describe what to build next
Auto



schedule effectively